Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

STUDY_AI

A Personalized AI-Powered Flashcard Generation & Vault Management System

Neelkamal Gupta
2210991969

Supervised By
Dr. Rajat Takkar
(Assistant Professor)

Department of Computer Science and Engineering,
Chitkara University, Punjab

Introduction

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The Problem I Solve:

CURRENT PAIN POINTS

- Generic AI chats produce ad hoc questions with no persistent organisation
- Traditional flashcard tools require learners to write questions manually
- No system personalises output to the student's academic level or degree
- Study photos and notes are not leveraged for automated question generation
- No tool provides metacognitive analytics on cognitive depth

STUDY AI BRIDGES THE GAP

- ✦ Two input modes: topic keyword + image OCR
- ✦ Profile-driven AI personalisation
- ✦ Persistent vault with practice mode
- ✦ Real-time Bloom's Taxonomy analytics
- ✦ Full security with JWT & bcrypt

OBJECTIVES

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Project Objectives

01 Convert study material into personalised, exam-ready flashcards via two input modes: free-text topic search & image OCR

02 Implement dynamic profile-driven prompt construction using 6 user parameters (age, degree, difficulty, question style, gender, goal)

03 Deliver an interactive flip-card UX grounded in active-recall pedagogy — answer hidden until deliberately revealed

04 Provide a persistent personal vault with search, sort, CSV/PDF export, and 3D-flip practice mode

05 Implement client-side Bloom's Taxonomy classifier and Flesch-Kincaid readability analyser for metacognitive feedback

06 Secure the platform with JWT, bcrypt, Helmet, rate limiting, and build an admin analytics dashboard

System Overview

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StudyAI converts any study material - typed topic or photographed page - into interactive Q&A flashcards personalised to each learner's academic profile.

Input

Topic text or Image

AI Engine

Gemini 1.5 Flash

Flashcards

Interactive 3D flip

Vault

Save & Practice

Analytics

Bloom's + Readability

Key Design Principle: Flashcard answers are HIDDEN by default - learners must actively flip to reveal, reinforcing retrieval practice and active recall.

Tools & Tech Used

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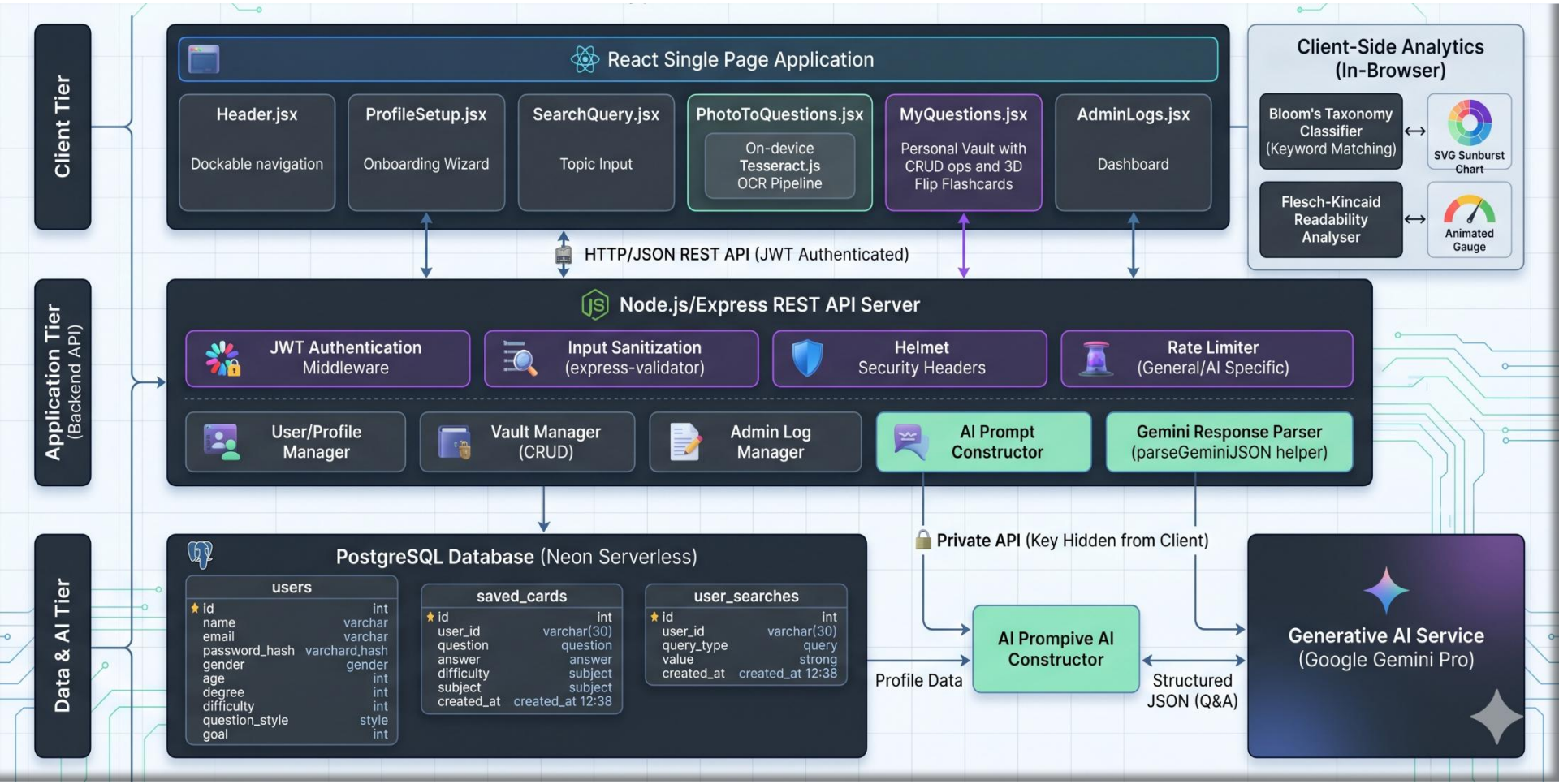
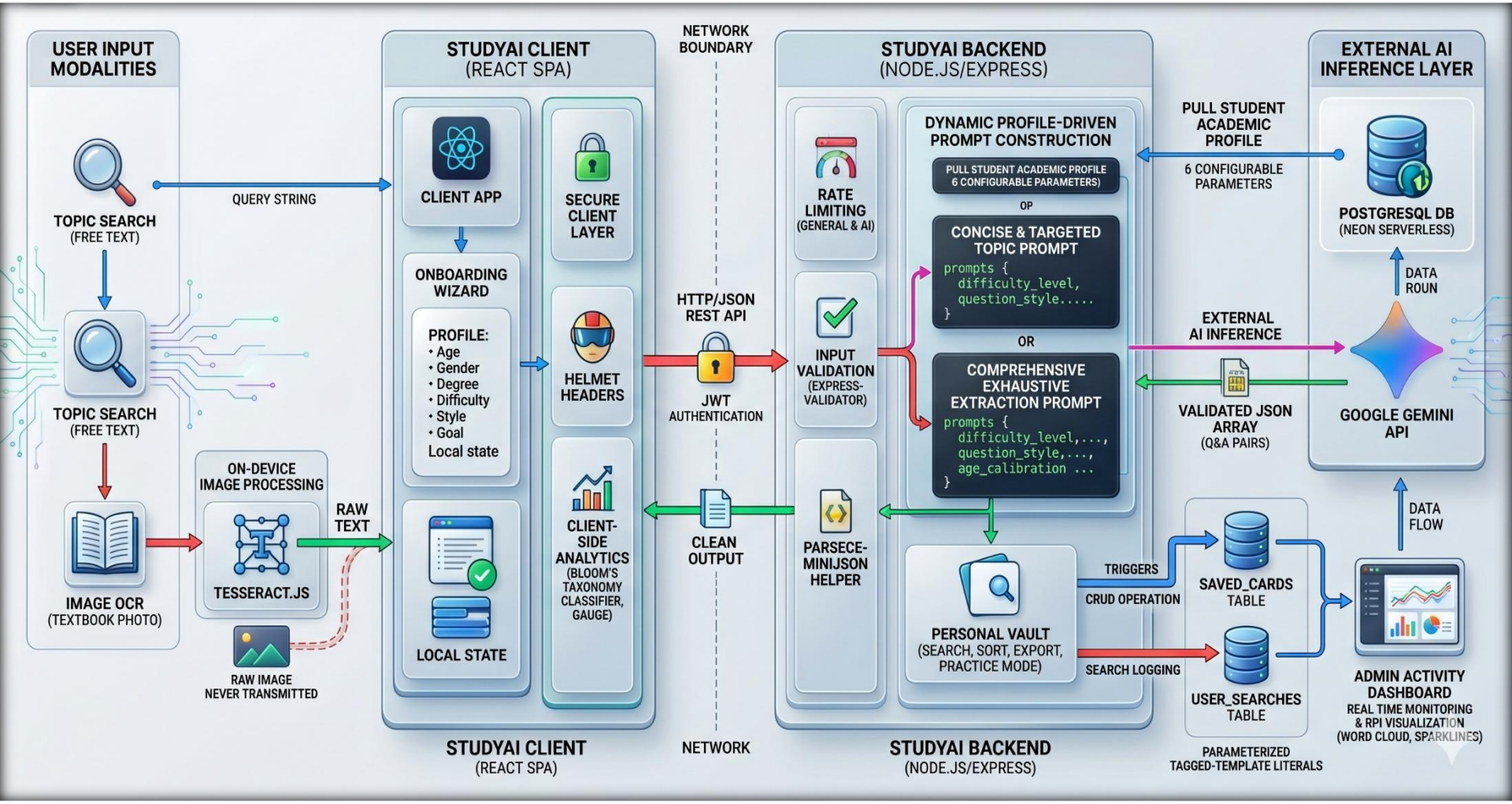
FRONTEND	BACKEND	AI & SERVICES
• React 18 + Vite (Component SPA) • Tailwind CSS + CSS-in-JS • React Router v6 • Tesseract.js (on-device OCR)	• Node.js + Express.js (REST API) • PostgreSQL via Neon (Serverless DB) • JWT + bcryptjs (Auth) • Helmet, express-rate-limit, express-validator	• Google Gemini 1.5 Flash (Q&A generation) • Dynamic profile-injected prompts • JSON-structured AI output • dotenv (secrets management)

Three-tier architecture: React SPA → Express REST API → Neon PostgreSQL + Gemini AI

System Architecture Diagram



Sequence Diagram



Key Features at a Glance

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Dual Input

Free-text topic search and client-side image OCR (Tesseract.js) — no image ever sent to server

Profile-Driven AI

6 user parameters (age, degree, difficulty, style, gender, goal) injected into every Gemini prompt

Interactive Flashcards

3D-flip cards with answer hidden by default; keyboard, swipe & click support; confidence self-rating

Personal Vault

Save, search, sort, delete, export (CSV & PDF), and practise saved cards with 3D-flip practice mode

Bloom's Analytics

Client-side 6-level cognitive classifier + SVG sunburst chart + Flesch-Kincaid readability score

Security

JWT + bcrypt, Helmet headers, rate limiting (10 req/min on AI endpoints), input validation, parameterised SQL

Security Implementation

Password Storage	bcryptjs (salt rounds = 10)
Session Management	JWT — 24-hour expiry, signed with env secret
HTTP Headers	Helmet.js — X-Frame-Options, CSP, Referrer-Policy
Rate Limiting	100 req / 15 min (global) • 10 req / 60 s (AI endpoints)
Input Validation	express-validator — sanitizes all fields; 400 on failure
SQL Injection	Neon tagged-template literals — zero string concatenation in SQL
Image Privacy	Tesseract.js runs on-device — no image ever leaves the browser
API Key Isolation	Gemini key & JWT secret server-only; never sent to client

Contributions

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I

Privacy-Preserving Client-Side OCR Pipeline

Tesseract.js runs entirely in the browser. The raw photograph is never transmitted to any server — Express, Gemini, or any third-party. Only extracted text travels over the network. A meaningful privacy guarantee for students photographing personal notes or institutional exam material.

II

Dynamic Profile-Driven Prompt Construction

Six independently configurable parameters (age, gender, degree, difficulty, question style, goal) stored in PostgreSQL are injected into every AI prompt. A 19-year-old CS undergraduate on 'intermediate' receives demonstrably different questions than a 28-year-old EE postgraduate on 'advanced' — for the identical topic.

III

Client-Side Bloom's Taxonomy Classifier + Real-Time Analytics

A keyword-matching classifier runs post-generation in the browser, tagging each question across 6 cognitive levels (Remember → Create). Rendered as an interactive SVG sunburst chart with per-level bar breakdown. No additional server call — zero latency, zero cost. Not found in any comparable open-source or commercial flashcard tool.

Future Scope & Planned Enhancements

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Spaced Repetition (SM-2)

Persist confidence ratings; surface due-for-review cards via SuperMemo SM-2 forgetting-curve algorithm

RAG Pipeline

Upload full PDF courses; Retrieval-Augmented Generation grounds answers in the specific document

Multi-Modal Input

Handwriting recognition, diagram annotation, and table extraction from scanned images

Collaborative Vaults

Shared flashcard decks — publish, fork, and collaboratively annotate with a study group

Containerised Deployment

Docker + Cloud Run with connection pooling and CDN-hosted static assets

Multi-Language OCR

Extend Tesseract.js to regional Indian languages: Hindi, Tamil, Telugu

Conclusion

- StudyAI is a full-stack, AI-integrated web application that successfully bridges the gap between raw study material and structured, practice-ready flashcards. The system's two input pathways - free-text topic search and on-device image OCR - cover the most common sources of student study material, while the profile-driven prompt construction ensures that generated content is calibrated to each learner's academic level and preferred cognitive style.
- The core interaction model - interactive flip flashcards where the answer is hidden until deliberately revealed - is grounded in established active-recall techniques and is accurately represented throughout the application. The system does not position itself as a quiz engine; it is a flashcard generation and management platform with enriched analytics.
- The three original contributions - privacy-preserving client-side OCR, dynamic profile-driven prompt construction, and client-side Bloom's Taxonomy classification - collectively constitute a novel integrated framework that has not been reproduced in existing academic or commercial tooling. The modular, iterative development approach, evidenced by the retained version history in the codebase, demonstrates structured software engineering practice. The project is ready for peer review, continued academic use, and incremental extension toward production-grade deployment.

Thank You